



# AI-Driven Prediction of Sugarcane Quality Attributes Using Satellite Imagery

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## Abstract

Anticipating the sugar content of sugarcane crop is a crucial aspect that holds the key to develop innovative data-driven solutions for determining the ideal time of mechanized harvest. However, traditional laboratory-based approaches are laborious, time-consuming, and limited in their scalability. Thus, we explored the potential of integrating multispectral data and cutting-edge artificial intelligence algorithms to predict the sugar content attributes of sugarcane, namely Brix and Purity. The sugarcane quality attributes were measured in a routine laboratory using 510 georeferenced samples from two commercial areas. Crop canopy reflectance values and growing degree days (GDD) were used as inputs on developing predictive models. Two artificial intelligence (AI) algorithms, artificial neural network (ANN) and random forest (RF), and multiple linear regression (MLR) were performed to create the predictive models for mapping the sugarcane quality. The models' performance proved that RF regression was better for °Brix prediction. In contrast, Purity values were better predicted by ANN algorithm. GDD was the most important variable on performance of RF modeling for both outputs, followed by Green spectral band from satellite imagery. Timely results were achieved integrating satellite imagery and AI-based model on prediction of qualitative attributes for sugarcane. It can provide useful data layers to support site-specific management strategies within season by agroindustry and supporting the decision-making of harvesting in large scale.

**Keywords** Brix · Crop monitoring · Machine learning · *Saccharum* spp

## Introduction

The diversified remote sensing applications in agriculture have enable a higher capacity of data acquisition in a shorter interval of time in comparison with traditional methods of sampling. The data-driven solutions, integrated with mechanized operations on agricultural areas, allowed to optimize crop yield and quality, the use of resources, and have enabled a more sustainable agriculture (Žibrat et al. 2019). Bolfe et al. (2020) conducted one national survey about the digital agriculture adoption in Brazil. Their main outcomes were that 84% of the interviewed growers adopted at least one digital tool on farms, and about 20% of those growers use some type of remote sensing tool for a previous diagnosis

of the field, mainly, for detecting plant diseases and pests, hydric stress, and weed management.

Sugarcane (*Saccharum* spp.) production chain can be benefited by innovative solutions that estimate crop yield and quality as a guide for adopting precision agriculture strategies. Some examples are local interventions on crop growth and logistical planning of harvesting. The mechanized harvest of sugarcane is the major operational cost for suppliers that represents around 33% of the total cost of production on industry (Martins et al. 2023). The traditional method of payment to the sugarcane growers considers the total mass of raw material and ATR (total recoverable sugar), which is an indicator of the crop quality. The determination of ATR depends on the concentration of soluble solid content (°Brix), sugar content (Pol), and fiber of the stalks. Those variables are commonly measured in laboratory by spectrometry (Phuphaphud et al. 2019) using destructive samples with a high cost of analysis, besides being time-consuming and generating low data resolution, which isn't

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useful to design local interventions at the field in large scale (Corrêdo et al. 2021).

For sugarcane production systems, it is relevant to monitor the net photosynthesis and available sucrose in crop stalk, which decreases near maturity stages, to determine the ideal moment for harvesting the raw material considering its cost-effectiveness. Since chlorophyll pigments are an indicator of the physiological conditions of the crop, those variables are commonly measured indirectly by crop canopy reflectance using multispectral data from satellite imagery or unmanned aerial vehicle (UAV) (Oliveira et al. 2022). Consequently, chlorophyll pigment could serve as a promising indicator of sugar content. Despite the importance of such remote sensing technologies in predicting sugar content in sugarcane, limited studies focused on this area. For instance, Chea et al. (2020) used multispectral data and a linear regression model to analyze °Brix, Pol, fiber, and commercial cane sugar (CCS). Their results showed varying levels of accuracy, ranging from weak to good ( $R^2 = 0.35\text{--}0.84$ ) relationship. Similarly, Todd et al. (2022) investigated RGB spectral indices to establish a correlation with Brix in a sugarcane breeding field. Their results showed a moderate correlation ( $r = \sim 0.6$ ). However, although the promising results demonstrated by these studies, statistical capabilities are limited by relying on linear regression or correlation, which hinders the realistic modeling of the field. Therefore, it is imperative to invest in sophisticated data analysis techniques and orbital platforms to address these limitations. Chea et al. (2022) conducted a study that investigated the application of machine learning models for Brix prediction using multiple data resource, such as multispectral imagery, crop height/age, and weather data. Although robust, their approach reached moderate accuracy ( $R^2 = 0.54\text{--}0.75$ ). Furthermore, the use of multiple types of data sources presents challenges, especially in regions where such data is scarce. Finally, the studies have primarily focused on aerial remote sensing platforms, which poses scalability issues, thus further emphasizing the need for more comprehensive research in this area.

The use of AI enables the prediction of both quantitative and qualitative characteristics of the crops. AI algorithms are particularly adept at addressing the complexity inherent in datasets and accounting for the considerable variability in observations. The scientific community has recognized the adoption of AI-based methods for agricultural applications as an alternative manner of mapping complex landscapes (Wang et al. 2019), predicting crop yield (Wei et al. 2020), and improving mechanized grain seeding through automated adjustments (Sheikh et al. 2021). Particularly, artificial neural network (ANN) and random forest (RF) algorithms are commonly used AI techniques to process remote sensing data and extract relevant information to agricultural context. For instance, ANN modeling had been successfully applied to estimate dry biomass and canopy diameter in soybean

(Carneiro et al. 2022). Conversely, the RF algorithm has been used to predict above-ground biomass of wheat (Yue et al. 2018) and corn (Schwalbert et al. 2020).

Therefore, it is widely recognized in the literature the prospective implications associated with the integration of crop canopy reflectance, agronomic data, and AI-based models to address intricate complex challenges in tropical agriculture. Consequently, our objectives are to evaluate the effectiveness of crop canopy reflectance from in-season satellite imagery to develop predictive models related to the sugarcane quality attributes. These predictive models were developed using both traditional approaches, specifically multiple linear regression (MLR), and AI-based methods.

## Materials and Methods

The workflow of the methodology was divided into data collection, pre-processing data, fitting the models, and assessment of the model's performance. It includes each step from data collection to the development of predictive models for qualitative attributes of sugarcane integrating high-resolution imagery data, climatic data, and the georeferenced measurements at the field.

### Data Collection

The study was carried out in two commercial sugarcane fields located near the city of Jaboticabal, São Paulo, Brazil ( $21^\circ 18' 55''\text{S}$ ,  $48^\circ 9' 35''\text{W}$ ; altitude 562 m above sea level), during two crop growing seasons. The crop variety was planted on a row spacing of 1.50 m for both fields. The sugarcane varieties are considered ideal for the selected areas and widely used by the local growers (Cursi et al. 2022). Both varieties are characterized by high yield (about  $86 \text{ t ha}^{-1}$ ) and high saccharose content, late-maturing, and favorable production on environments A–C. The fields were mechanically green harvested (no burn) during mid-season (dry period). The region has Oxisol soil type (typical dystrophic red latosol), and it is composed mostly by clay content as described by Montanari et al. (2010). The climate is Aw with a summer dry season. Annually, rainfall reaches about 1460 mm and the average temperature is  $22.6^\circ\text{C}$ . The characteristics of both fields and crop conditions are shown in Table 1. Plant cane or younger ratoons weren't considered on study due to the selection of the areas according to its availability by the growers and operational capacity of collecting samples manually on both study sites.

Eight samplings of sugarcane stalks were performed manually in Field 1 from tillering stage to the crop maturity stage with an interval between samplings of 15 days. For each analysis, the georeferenced data were collected at 30 sample points regularly distributed ( $9 \times 9 \text{ m}$  grid),

**Table 1** Information of the evaluated sugarcane fields

| Field | Area (ha) | Season | Variety    | Ratoon | Number of samples |
|-------|-----------|--------|------------|--------|-------------------|
| 1     | 0.37      | 2022   | RB 97–5201 | 6th    | 240               |
| 2     | 17.86     | 2023   | CTC 20     | 8th    | 270               |

total of 240 samples (30 samples  $\times$  8 dates). For Field 2, the georeferenced data were collected at 90 points of sampling regularly distributed (40  $\times$  40 m grid) from tillering stage to the crop maturity stage once a month, total of 270 samples (90 samples  $\times$  3 dates). The smaller number of samples collected on Field 2 is due to the larger area involved on study.

Therefore, a total of 510 samples were collected at the field and analyzed in laboratory to measure the Brix and Purity contents of sugarcane. The whole stalks were manually cut at the ground level. The growing point and the leaves were removed before sending those samples for a hydraulic press. The collected sugarcane stalks were processed and analyzed individually by date of sampling in a routine laboratory for measuring Brix and Purity contents according to the conventional protocol proposed by the National Council of Sugarcane Producers of São Paulo State (CONSECANA 2015). The protocol agrees with the International Commission for Uniform Methods of Sugar Analysis.

For the traditional analysis of crop qualitative attributes, each 500 g sample of stalks was pressed in a hydraulic press under constant pressure (24.5 MPa) for 60 s to obtain the raw juice. The Brix content was measured for each sample by a digital refractometer (ABBE, Atago Pal-1, Tokyo, Japan) with a maximum resolution of 0.1°Brix. After measuring °Brix, 12 g of a mixture composed of 4:2:1 proportion of Celite, aluminum chloride, and calcium hydroxide was added to 200 ml of raw juice and homogenized using the agitation process by a magnetic stirrer. Then, the solution was filtered by means a paper filter to obtain a transparent solution. A volume of 100 ml of the filtered solution was used for saccharimeter measuring by a digital polarimeter (Anton Paar, Bremen, Germany) to determine the Purity content of the samples.

The growing degree days (GDD—Eq. 1) was included as an input variable on predictive models due its relationship with sugarcane phenology. The basal temperature of the crop was considered as 18 °C (Bachi and Souza 1978). The weather data were collected by means an automatic weather station (Davis Instruments, equipment Vantage Pro Plus Wireless), located around 10 km from the evaluated sugarcane fields, considering the daily temperature correspondent to the period of crop development for both seasons.

$$\text{GDD} = \left( \frac{T_{\max} + T_{\min}}{2} \right) - T^b \quad (1)$$

where GDD is the growing degree days;  $T_{\max}$  is the maximum air temperature (°C);  $T_{\min}$  is the minimum air temperature (°C); and  $T^b$  is the basal temperature of the crop (°C).

## Imagery Data

Orbital images from nanosatellites PlanetScope were utilized from the same period that data were collected during each crop growing season for Field 1 (eight images) and Field 2 (three images). The constellation uses a multi-spectral instrument (eight spectral channels) with a 25 km swath width and a ground sample distance (GSD) of 3.0 m. All images were downloaded from Planet Explorer website by means of Education and Research Program of Planet Labs. The original images are orthorectified and scaled to the top of atmosphere radiance internally on-board sensor. Just the images with low cloud cover (< 1%) were selected for image processing and clipped according to the field boundaries. All imagery data and vector layers were projected on WGS/UTM zone 22S using Quantum Geographic Information System (QGIS 2.18.26). An internal buffer of 5.0 m was applied to the field boundaries to ensure that the spectral data corresponded to the experimental site of sugarcane plants. The individual reflectance values of four spectral bands: Red (590–670 nm), Green (500–590 nm), Blue (455–515 nm), and NIR (near-infrared, 780–860 nm) from satellite imagery were considered in the study to avoid overfitting of the predictive models and due its citation for other prediction analysis in agriculture (Canata et al. 2021). Each in-season phenological stage of sugarcane, from tillering to maturity crop stage, was identified according to the days after cutting (DAC) considering the most recent harvest before starting data collection in the field.

## Pre-Processing Data and Training Models

The original dataset was composed of 510 samples. A total of five independent variables were considered to perform modeling: reflectance values from spectral bands (four variables) and GDD. The qualitative attributes (Brix and Purity) were the dependent variables to be predicted individually. All input and output parameters were grouped and normalized between 0 and 1 values according to Eqs. 2 and 3.

$$x'_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (2)$$

$$y'_i = \frac{y_i - y_{\min}}{y_{\max} - y_{\min}} \quad (3)$$

where  $x_i$  represents each input variable;  $y_i$  is the predicted value;  $x_{\min}$ ,  $x_{\max}$ ,  $y_{\min}$ ,  $y_{\max}$  represent their corresponding minimum and maximum values;  $x'_i$  and  $y'_i$  are normalized input and output parameters, respectively.

All statistical analyses and development of the predictive models were performed using the R programming language (version 4.2.2). An additional exploratory analysis was the correlation between the variables selected from previous step and the output values using Pearson's correlation coefficient. The normalized dataset was randomly split into training and test data considering a ratio of 70:30. The train data were used to carry out the training of each predictive model developed in this study.

The attributes measured in the field were interpolated individually by means of IDW (Inverse Distance Weighting) method on QGIS 2.18.26 as no spatial dependence was observed on geostatistical analysis of the georeferenced dataset. The spatial resolution on interpolation was set equal to 3.0 m due to the same spatial resolution of the satellite imagery. The relationship between measured and predicted values of the sugarcane quality attributes was assessed by visual comparison of the maps and using the map of prediction error by attribute.

The selection of variables is an important step of data processing to perform predictive models that aims to find a pattern between the features (variables) and performing the most suitable modeling to classify or predict the output values. The iterations are continuously done until the model to achieve a minimal error among the observed value and the predicted value (Morota et al. 2018). The open-source package *olsrr* (Neter et al. 1996) was carried out to select the best features for predicting Brix and Purity contents. The regression model with the highest coefficient of determination ( $R^2$ ) and the lowest mean squared error of prediction (MSEP) was considered to select the input variables for prediction by MLR and ANN methods.

The variable importance on RF regression was determined using all predictors and fitting the model after the permutation of each predictor variable (*varImp* function). All the important measurements were scaled, ranging from zero to one, which determined the mean decrease accuracy (%IncMSE) and mean decrease impurity (IncNodePurity) of the fitted model. This procedure was applied considering that supervised and unsupervised methods, such as RF regression, tend to improve the prediction with the increase in predictors. So, it is necessary to discern its complexity to avoid an underfitting or overfitting of the results from modeling (Lindner et al. 2022).

## Development of the Predictive Models

The regression methods MLR, RF, and ANN were performed to predict Brix and Purity contents as it enables to achieve

results with high accuracy for many input features. The function *set.seed* was used with the value 1234 before fitting each predictive model. MLR was built-in using the function *lm* and RF regression from *randomForest* package (Liaw and Wiener 2002). ANN was built by means of *neuralnet* package (Günther and Fritsch 2010). MLR is a statistical modeling technique that assumes a linear relationship of the variables. It offers several advantages, such as its simplicity and interpretability, making it a popular choice for analyzing complex datasets. Furthermore, MLR allows the inclusion of multiple independent variables, which enables the exploration of their collective influence on the dependent variable.

RF regression is a combination of decision trees that each tree depends on the values of a random vector sampled independently from input vector. For each tree, the data are recursively split into more homogenous units, namely nodes. To improve the predictability of the response variable on RF regression, it was necessary to fine-tune two parameters: (i) *ntree*: the average result of many trees and (ii) *mtry*: the predictor subset value. To find the best *ntree* value, the prediction error rate was calculated for a range of 0–500, and the one that presented the lowest error before stabilization was selected (*elbow rule*). The *ntree* was set as 200 to fit the model based on RF regression. The predictor subset was restricted to a minimum size of two variables (*mtry*=2). The *mtry* selection was the one that presented the lowest root mean square error (RMSE) on RF regression model.

ANN is a sub-field of artificial intelligence designed according to the relationship between the variables of the dataset. Three layers are required for data processing by ANN: input, hidden, and output layers. The input layer contained the number of neurons equal to the number of input variables selected previously. The number of hidden neurons was selected according to the smallest error generated between the observed data and the output value by ANN (*hidden* = 5000). The output layer contained one neuron that is the predicted value (Brix or Purity). Considering a feed-forward back-propagating ANN structure, used in this study, the neurons from one layer are connected to the neurons in the next layer and the information flows forward from the input to the output through the hidden layers (Kamalov et al. 2021). A neuron accepts the data from the previous layer and calculates a weighted sum of all its inputs (Eq. 4). Then, the transfer function is applied to the weighted value to calculate the output (Eq. 5).

$$t_i = \sum_{j=1}^n w_{ij} x_j \quad (4)$$

$$o_i = f(t_i) \quad (5)$$

where  $t_i$  is the weighted sum of all its inputs;  $n$  is the number of inputs;  $w$  is the weight of the connection between neurons

$i$  and  $j$ ; and  $x$  is the input from neuron  $j$ .  $o_i$  is the neuron output;  $f(t_i)$  is a transfer function applied to the weighted value  $t$  of inputs.

The sigmoidal function was used for hidden and output layers, and the linear transfer function was used for input layer as recommended for agricultural applications involving artificial intelligence (Maldaner et al. 2021). The accuracy and precision of the prediction by each fitted model were verified comparing the observed data and the predicted values. The error of prediction, for each sampled point, was determined by the difference among the observed data and the predicted value of Brix content by MLR, RF, and ANN. The same procedure was used to measure the prediction error of Purity content. All models were compared considering the metrics of modeling performance: RMSE,  $R^2$ , mean absolute percent error (MAPE), and mean absolute error (MAE). The regression model is considered better with values of  $R^2$  near to 1 and error values closer to 0.

## Results and Discussion

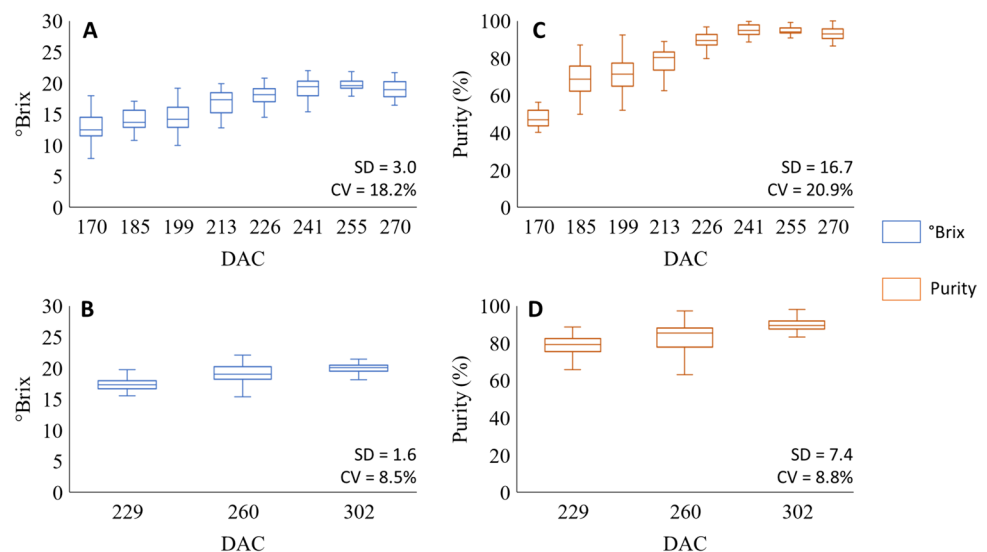
The original dataset for Brix content ranged from 7 to 20° along the development cycle of the crop for Field 1 (Fig. 1a) and from 14 to 20° for Field 2 (Fig. 1b). Both fields were observed a tendency to decline the values of °Brix on maturation crop stage. The values of Purity ranged from 40 to 99% for Field 1 (Fig. 1c) and from 80 to 99% for Field 2 (Fig. 1d). The higher minimum values of °Brix and Purity for Field 2 are explained by the fact that data collection started later compared to the Field 1, so, there was a higher cumulative of sugar content on stalks for the conditions on Field 2. Additionally, °Brix and Purity contents had about 20% coefficient of variation (CV) on Field 1, which characterize low-moderate spatial variability of the attributes

(Gomes and Garcia 2002). Considering the same criteria, those attributes demonstrated a low spatial variability on Field 2 (CV < 10%).

The understanding of crop quality variability can guide the identification of the limiting factors on crop response and the delimitation of site-specific management practices to optimize the agricultural production in large scale. A homogeneous distribution and trend of the assessed parameters were verified on dataset along the time interval of evaluation. The highest values for Brix and Purity were achieved at 7th evaluation of sugarcane sampling on Field 1, which indicates the most appropriate moment for sugarcane harvesting due to the high-quality material for industry. The phenological crop stage is an important criterion to establish the site-specific management, so, according to the results of this study, the suggested moment for sugarcane harvesting coincided with the transition period between the development and ripening stage of the crop (about 270 DAC associated with the initial process of decline in saccharose). The direct or indirect measurements of crop quality, along the phenological stages, support to differentiate agricultural areas where quality parameters of the raw material are below the desired standards for industry. Thus, its spatialized differentiation is a useful information for planning the logistics of mechanized operations and the local interventions that aims to improve the crop profitability and sustainability over the seasons.

The regression analysis integrating multispectral data and a crop growing indicator, such as GDD, pointed out the best features for predicting Brix and Purity contents of sugarcane. The prediction of °Brix was based on five input variables (Blue, Green, Red, NIR, and GDD) with  $R^2 = 0.69$  and  $MSEP = 2.77^\circ\text{Brix}$ . The prediction of Purity considered four input variables (Blue, Green, NIR, and GDD) with  $R^2 = 0.79$  and  $MSEP = 2.93\%$ . In this case, the same level of variable

**Fig. 1** Descriptive statistics of °Brix on Field 1 (A), Field 2 (B) and Purity on Field 1 (C), Field 2 (D)





importance was considered for predictive models by ANN and MLR methods. For RF regression, the variable importance was calculated over the cited variables for each output.

The correlation between the variables for predicting °Brix and Purity contents is shown in Table 2. The reflectance value of blue spectral band from satellite imagery was the most correlated variable with °Brix ( $r=0.69$ ) and Purity ( $r=0.54$ ). A moderate relationship was verified between °Brix and GDD ( $r=0.54$ ). NIR spectral band has exhibited significant correlations with sugar content, as evidenced in previous studies to evaluate sugar content of sugar beet roots (Wang et al. 2022), as well as when using chlorophyll index rededge (CIrededge) to assess sugar content in sugarcane crop (Chea et al. 2020). But, on this study, no correlation

was observed between the predicted variables and NIR spectral band.

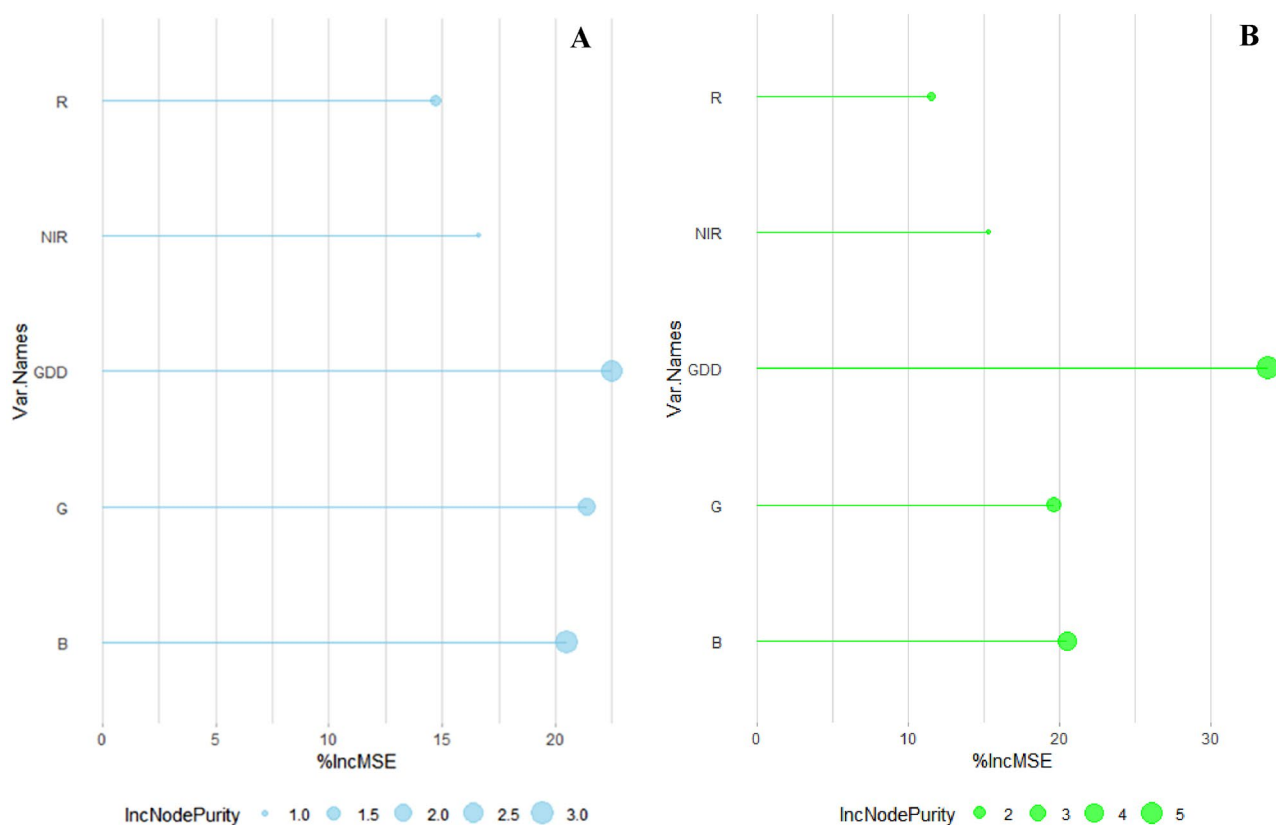
The exploratory analysis conducted on the dataset indicated a high correlation between the Blue spectral band and the qualitative attributes of sugarcane. The observed behavior of the spectral feature presents a dynamic challenge when attempting to conduct a reliable analysis using MLR method, as it relies entirely on the assumption of a linear relationship between dependent and independent variables. Thus, it is essential that integrating these data into a nonlinear model can lead to improved performance. This proposition finds support in previous research, where AI models have proven successful on predicting sugar content of beet root (Wang et al. 2022), grapes berries (Ryckewaert et al. 2023), and sweet sorghum (Simeone et al. 2017). Thus, integrating in-season multispectral data and artificial intelligence methods provide an alternative to improve the accuracy on prediction of °Brix and Purity contents for sugarcane.

The variable importance on RF regression is shown in Fig. 2. The three most important predictor variables for °Brix (Fig. 2a) and Purity (Fig. 2b) content were GDD, Green spectral band, and Blue spectral band from satellite imagery. In both cases, GDD was the variable with more importance on RF regression, as well identified in previous

**Table 2** Correlation between the input variables to predict Brix and Purity

|        | Red     | Green   | Blue    | NIR     | GDD     |
|--------|---------|---------|---------|---------|---------|
| Brix   | 0.51*** | 0.55*** | 0.69*** | 0.28*** | 0.54*** |
| Purity | 0.33*** | 0.35*** | 0.54*** | 0.12*** | 0.32*** |

\*\*\*  $p$ -value < 0.001; *NIR* near infrared; *GDD* growing degree days



**Fig. 2** Variable importance for random forest regression to predict °Brix (A) and Purity (B)

results for ANN and MLR methods. It demonstrates the importance of integrating phenological characteristics of the crop and the response of the crop (reflectance measurements) to the inherent conditions of the field. Red spectral band had less importance for predicting both outputs on RF regression.

The results demonstrated that a single spectral band isn't recommended for predicting qualitative attributes of sugarcane, since a better performance of the predictive model for °Brix and Purity values was composed by at least three predictors' variables. Siedliska et al. (2021) related the relationship between blue spectral band and sugar content of different crops using hyperspectral reflectance measurements. In this case, the reflectance values from blue region (455–515 nm) of the spectrum were used to discriminate the levels of phosphorus (P) fertilization within the field. Despite that the relationship is based on the influence of P on sugar assimilation of the crops, it isn't possible to confirm the same cause–effect relation on this study.

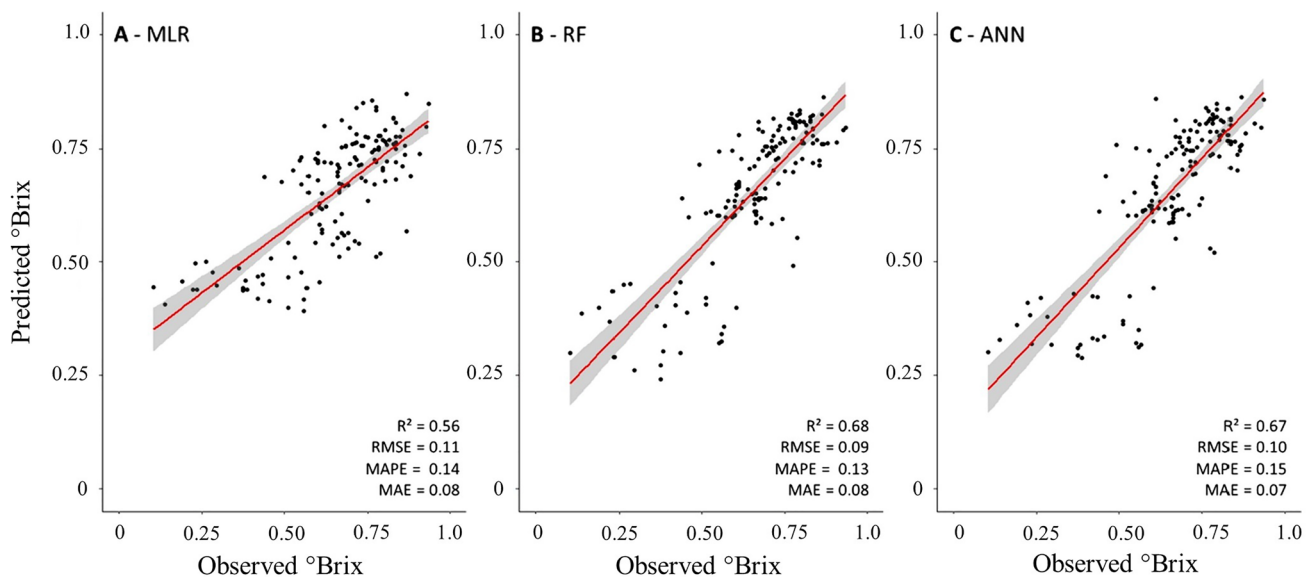
The results of the metrics to characterize the performance of each fitted model are summarized in Figs. 3 and 4. For both outputs, ANN, MLR, and RF methods performed very similarly in relation to the error of prediction (RMSE ranged from 0.09 to 0.11 °Brix, and from 0.11 to 0.18% Purity). The RF method demonstrated a better accuracy (Fig. 3b -  $R^2=0.68$ ;  $RMSE=0.09^\circ$ ) for predicting °Brix content than MLR and ANN algorithm. For Purity, a better accuracy was observed by ANN algorithm (Fig. 4c -  $R^2=0.74$ ;  $RMSE=0.11\%$ ).

Recent study demonstrated that the combination of RF algorithm and multispectral imagery optimized the biophysical modeling of sugarcane quality in relation to the methods

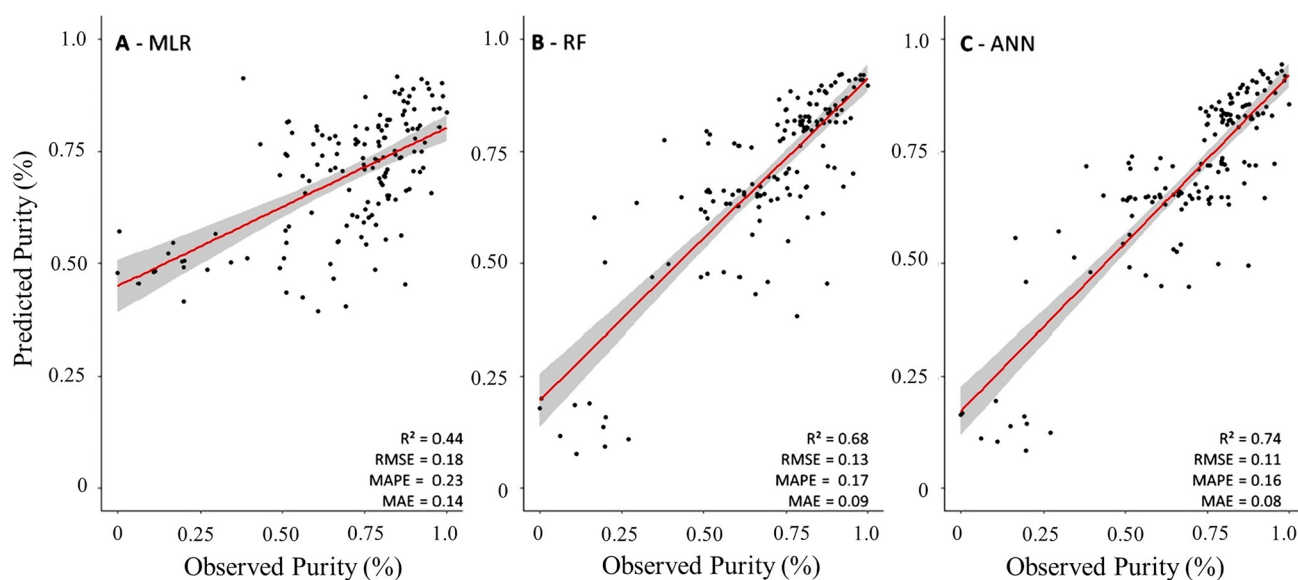
based on MLR, decision tree, and support vector machine (Barbosa Júnior et al. 2023). However, the performance of predictive models is highly dependent on the characteristics of the dataset. In cases where the dataset has a linear nature, linear models have the potential to outperform even robust models such as RF and ANN. Conversely, models such as ANN and RF models typically require a larger amount of data to effectively capture complex patterns and relationships. In cases where the dataset is limited in size, MLR models can demonstrate enhanced generalization capabilities and avoid overfitting issues that can arise when using more complex models, such as ANN and RF (Jiao et al. 2020).

The observed values of °Brix (Fig. 5a–b) and Purity (Fig. 5c–d) contents for both fields were mapped to identify the spatial variability of the qualitative attributes of sugarcane (ground-truth data) at the maturation crop stage. As verified on descriptive statistics of the dataset, there is a low variation in °Brix and Purity values on Field 2. Even so, the sugarcane quality can't be considered uniform for both fields, indicating that crop maturation varies over the commercial area, and therefore, the decision of harvesting the raw material should consider this variation.

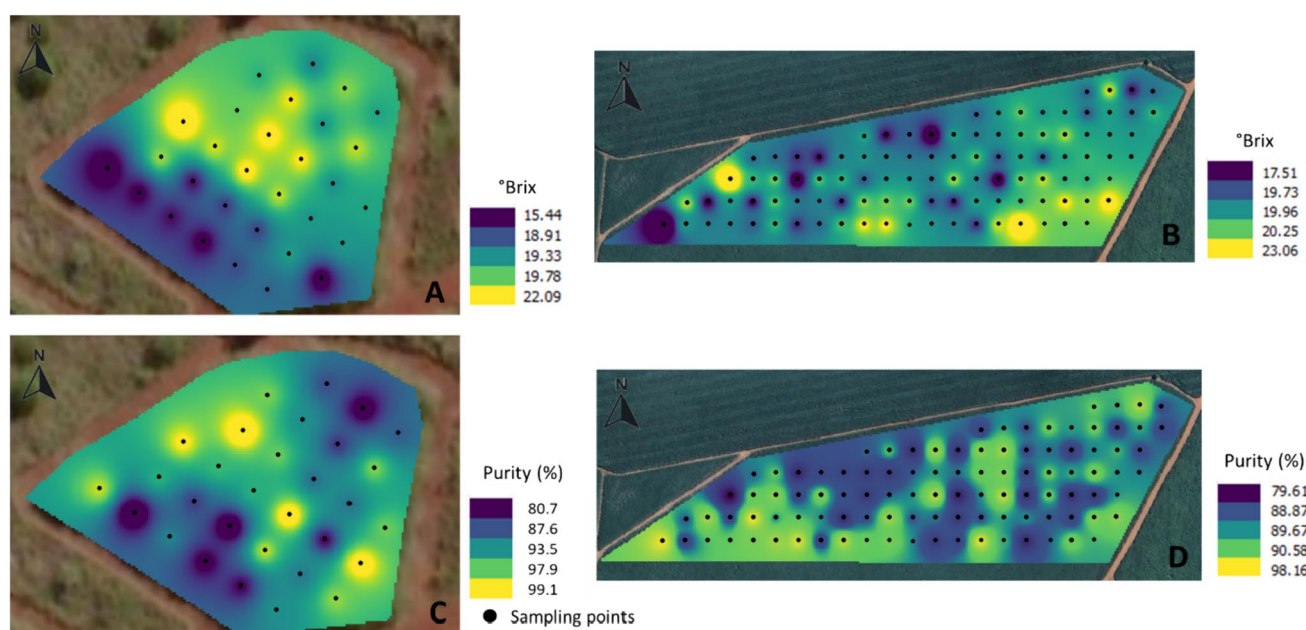
The relative error of prediction in relation to the observed values of °Brix (Fig. 6a–b) and Purity (Fig. 6c–d) contents indicated a consistency of the predictive models on estimating the qualitative attributes of sugarcane based on RF and ANN. The highest relative error ranged from  $-1.97$  to  $5.29\%$  on prediction of °Brix in both fields. RF regression underestimated the values of °Brix according to map analysis (relative errors below zero). In contrast, there is an overestimation of Purity values considering the predictive model based



**Fig. 3** Regression and metrics of performance by predictive model of °Brix content



**Fig. 4** Regression and metrics of performance by predictive model of purity content



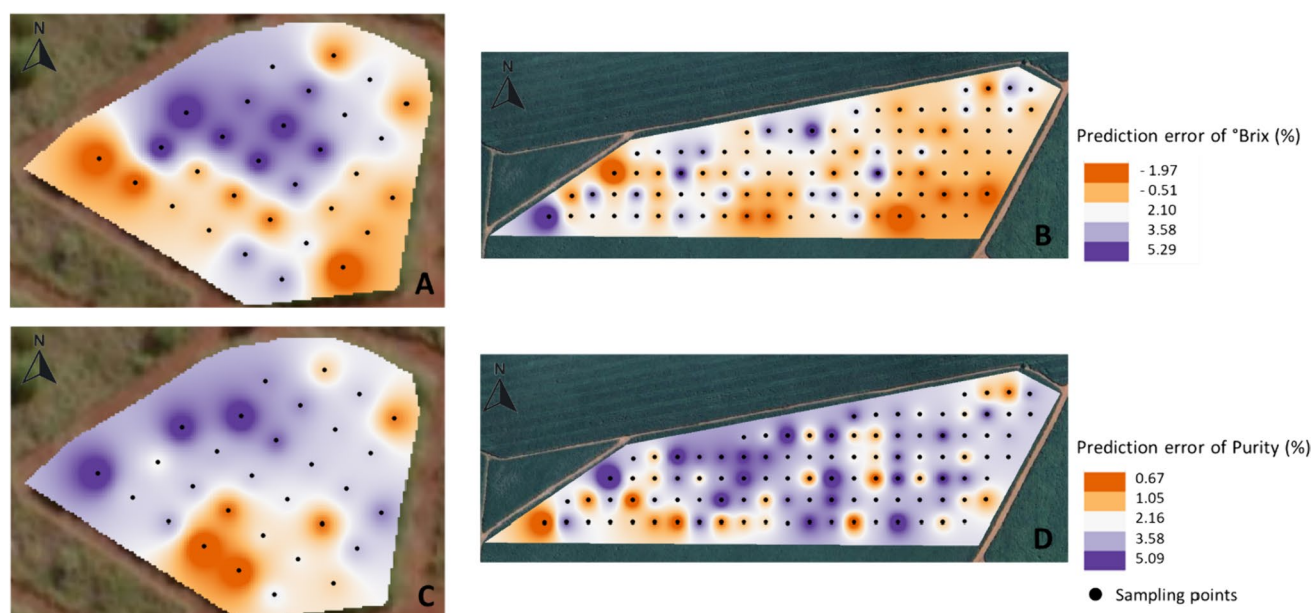
**Fig. 5** Maps of °Brix on Field 1 (A), Field 2 (B), and Purity on Field 1 (C), Field 2 (D)

on ANN algorithm, which showed a minor variation in the error to predict Purity content of sugarcane stalks (from 0.67 to 5.09%). The prediction errors about  $-0.51\%$  prevailed on estimative of °Brix and around 5% on prediction of Purity content for both fields, demonstrating the effectiveness of IA-based models on performance of prediction for qualitative attributes of sugarcane at the field level.

A reliable estimative of sugarcane quality can be applied on important approaches to make agriculture more

productive and sustainable. Since many factors can promote variability in the crop yield and quality (Zheng et al. 2022), it is expected to enable the identification of higher and lower potential sugarcane areas for harvesting at the optimal time based on crop mapping in advance of crop harvesting. Furthermore, the estimative can be used to identify the limiting factors of crop yield and quality for better plan its supply chain. Since the spatio-temporal variability of crop yield is an important information for the adoption of precision





**Fig. 6** Relative error of prediction for °Brix on Field 1 (A), Field 2 (B), and Purity on Field 1 (C), Field 2 (D)

agriculture practices, it should be interesting to perform integrated models for predicting for crop yield and quality as an alternative methodology to optimize agricultural resources.

Integrating high-resolution satellite imagery, GDD, and AI-based models demonstrated satisfactory accuracy on prediction of qualitative attributes for sugarcane crop. It can provide intensive crop mapping and useful layers of information to support precision agriculture strategies in commercial areas within season. Future studies should be developed to understand: i) the most appropriate in-season period to predict Brix and Purity contents based on multispectral data and vegetation indices; ii) the accuracy of its prediction considering different field conditions and crop varieties (e.g., drought-tolerant varieties). The technical viability of these approaches should be discussed in relation to the potential of improving the industrial process of sugarcane crop, such as to optimizing production of energy, food, and fuel in large scale.

## Conclusions

The developed predictive models based on RF regression demonstrated better performance to predict °Brix content of sugarcane ( $R^2=0.68$ ;  $MAE=0.08^\circ\text{Brix}$ ), and ANN algorithm showed better results for prediction of Purity crop content ( $R^2=0.74$ ;  $MAE=0.08\%$ ). The approach integrating multi-sources data, such as in-season satellite imagery and plant's phenology, demonstrated better outcomes for sugarcane quality prediction than analyzing individual features. GDD, a crop-growing indicator, was the most important

variable on performance of modeling for both outputs, followed by Green spectral band from satellite imagery. Thus, the development of predictive models provided spatialized information about qualitative attributes of sugarcane on tropical conditions, which is a potential indicator to optimize the schedule of mechanized harvesting by the agroindustry.

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**Data Availability** The database of this study is available on request from the corresponding author.

**Code Availability** Not applicable.

## Declarations

**Conflict of interest** The authors declare that there are no conflicts of interest regarding the publication of this paper.

**Ethical Approval** Not applicable.

**Consent to Participate** Not applicable.

Consent for Publication Not applicable.

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